Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/20/2005
PI Name:	Cowings, Patricia S. Ph.D.		
Project Title:	Development of an Index of Habitability Using Converging Indicators: Physiology, Performance, and Subjective Reports Physiology, Performance, and Subjective Reports		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Patricia.S.Cowings@nasa.gov	**Fax:**	FY 650-604-1484
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5724
Organization Name:	NASA Ames Research Center		
PI Address 1:	Human Systems Integration Division (TH)		
PI Address 2:	Mail Stop 262-2		
PI Web Page:	http://humansystems.arc.nasa.gov/		
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	06/30/2002	**End Date:**	12/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The purpose of this project is to develop and validate a quantitative method for assessing environmental effects on individual crewmembers during spaceflight. This project will involve analyses of data acquired during previous laboratory tests, field studies and spaceflight. NASA Ames Psychophysiological Research Laboratory has a unique database collected over a 25-year period on approximately 300 people exposed to a variety of environmental stressors, and the effects of both behavioral and pharmacological countermeasures on their stress tolerance. Previous research has shown that the use of converging indicators (physiological and subjective self-report measures) is a significantly more reliable method for assessing environmental effects on performance than any one indicator alone. We anticipate that the deliverable from this project will be a reliable methodology for accurately assessing microgravity effects on individual crew performance and for evaluating countermeasures designed to facilitate adaptation to space and readaptation to		

	Earth. Transfered database to Linux platform Developed Post processing software (PostProc) Establish protocols/methods for remote monitoring and training human physiological responses Data-base of biomedical and performance data from multiple experiments can be accessed from any PC platform Biomedical data can be processed, artifact removed and information derieved (e.g., cardiac output) from any PC platform Remote monitoring allows investigators from many different geographical locations to observe experiments in real-time. Remote training allows helath care providers to control physiological displays used for modifying these paramters (e.g., treatment of motion sickness, hypertension).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. Pilot (military, private. Search) training/airsickness abatement and performance enhancement. 2. Assessment and correction of environmental stressors effects on human health and performance. 3. Effective treatment for nausea and low blood pressure in various patient populations.
Task Progress:	*All objectives completed. Remote training tests in progress *Publications in peer reviewed journals during this grant period are: Kornilova, L.M., Cowings, P.S., Arlaschenko, N.I., Korneev, D.Y., Sagalovich, S.V., Sarantseva, A.V., Toscano, W.B., and Kozlovskaya, I.B. (2003) Individual characteristics of correction of the Cosmonauts' vegetative status with a method of adaptive biofeedback. Aviakosm Ekolog Med. 37(1), 67-72. *Papers in review (peer-reviewed journals) International Journal of Aviation Psychology entitled 'Autogenic Feedback Training Exercise: A Treatment for Air Sickness in Military Pilots.' 2001. Journal of Human Performance in Extreme Environments entitled 'San Francisco International Airport Tower Operations Involving Simultaneous Offset Instrument Approaches: Local Controller Workload Study.'
Bibliography Type:	Description: (Last Updated: 11/08/2023)
Articles in Peer-reviewed Journals	Kornilova LN, Cowings P, Arlashchenko NI, Korneev DIu, Sagalovich SV, Sarantseva AV, Toscano W, Kozlovskaia IB. "[Individual characteristics of correction of the cosmonauts' vegetative status with a method of adaptive biofeedback]" Aviakosm Ekolog Med. 2003;37(1):67-72. Russian. PMID: 12696507 , Jan-2003
Awards	Cowings, P.S.; Toscano, W.B. "NASA Space Act Award. June 2003." Jun-2003
Presentation	Toscano, W.B.; Cowings, P.S.; DeRoshia, C., Taylor, B.; Hines, S.A. "Individual Differences in Adaptational Capacity During Sustained Hypergravity" . 14th IAA Humans in Space Symposium, Banff, Alberta, Canada May-2003
Presentation	Cowings,P.S.; Toscano, W.B. "Psychophysiology of Spacefight" 14th IAA Humans in Space Symposium, Banff, Alberta, Canada May-2003
Presentation	Cowings, P.S., Toscano, W.B., Hines, A.S.; Jemison, M.; Ofili,E.; Lapubula, R. "Telemedicine Applications of Autogenic-Feedback Training Exercise" Research Programs of Historially Black Colleges and Universities, NASA, ARC Mar-2003
Presentation	Cowings, P.S.; Toscano, W.B. "Autogenic feedback Training Exercise Applications in Medicine and the Military" Virtual Reality Behavioral Health Meeting, Tripler Army Medical Center Honolulu, Hawaii, Oct-2003